

ASSEMBLYAI VOICE AGENT HACKATHON

LabLog

Run experiments with your hands.

Document them with your voice.

A voice-native AI laboratory notebook
& experiment copilot.



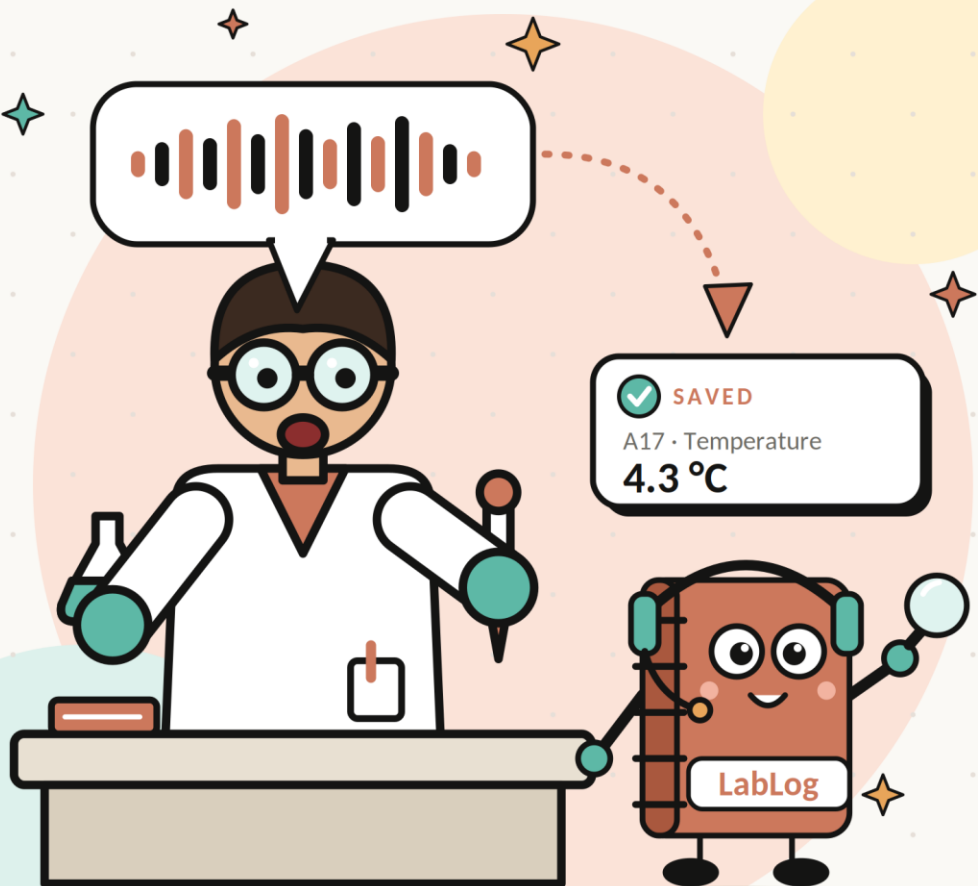
Hands-free



Validated records



Live dashboard



Built on the AssemblyAI Voice Agent API

THE MOMENT

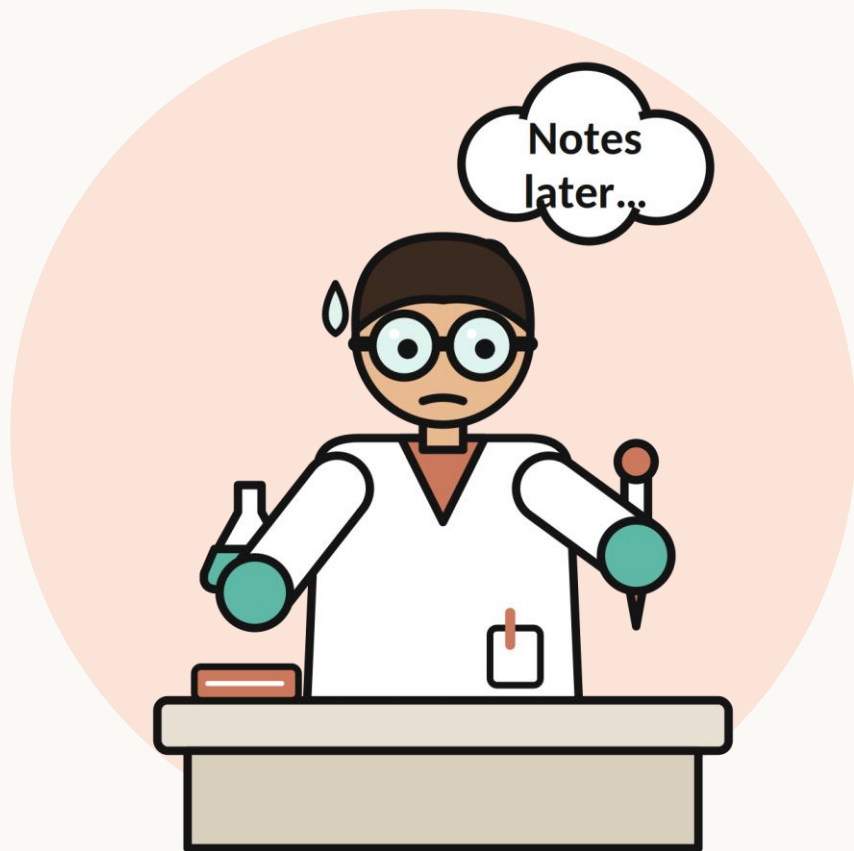
Hands full.
Gloves on.
Timer running.

And the notebook is across the room. So the numbers wait, and the details fade.

Times get rounded

Details get lost

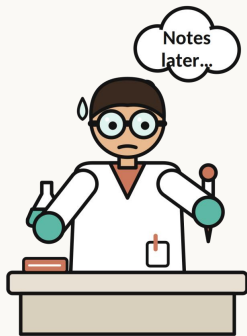
Values get retyped



TODAY

Write it down later.

- Stop, strip gloves, type
- Notes copied from scraps
- Audio memos nobody reads



WITH LABLOG

Say it now.

- Keep working, keep talking
- Checked and saved instantly
- A structured record with history



It doesn't just transcribe. It asks, and it acts.

WHAT YOU HEAR

"A18 is 4.1."

"Is that 4.1 degrees Celsius?"

"Yes."

"A18 recorded at 4.1 degrees Celsius."

"Actually, change A17 to 4.3."

WHAT LABLOG DOES

Unit missing. Nothing saved yet.

Asks instead of guessing.

```
record_measurement · validated · saved
```

Dashboard updates. Audit event logged.

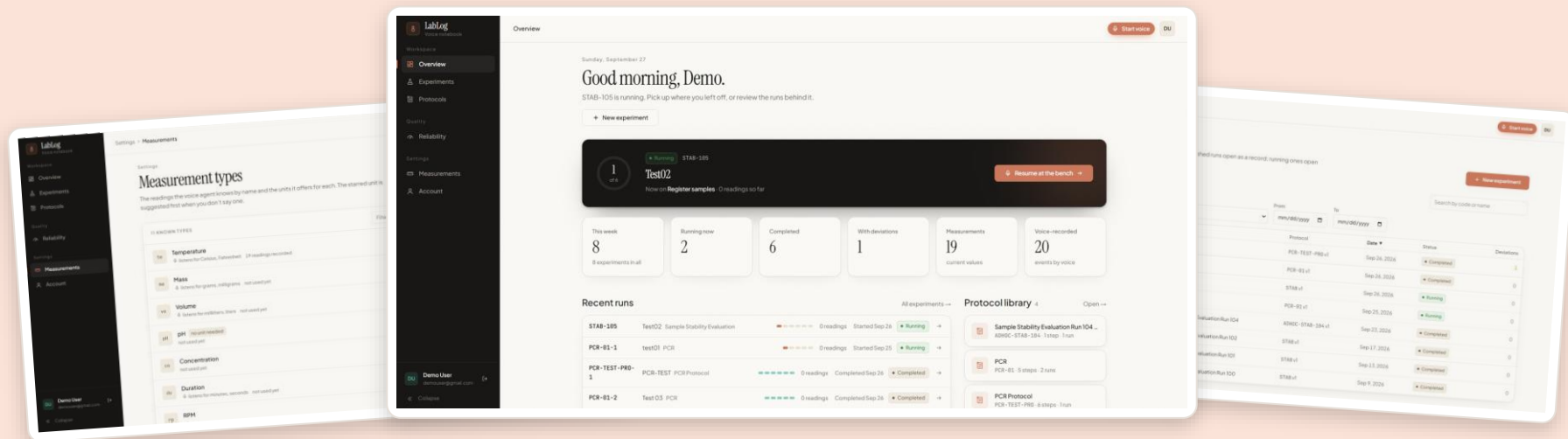
```
correct_measurement · 4.2 kept as previous
```

Not a voice memo. Not a chatbot.

	Voice memo	Typical voice chatbot	LabLog
Hands-free	✓	✓	✓
Structured records	—	—	✓
Follows your protocol	—	—	✓
Validates before saving	—	—	✓
Asks when unsure	—	Partly	✓
Keeps corrections and history	—	—	✓

Illustrative comparison of typical tools.

A working product, not a mockup.



8 experiments

19 measurements

20 voice-recorded events

demo workspace

Three things LabLog refuses to do.

No guessing.

YOU SAY

"Temperature is 37."

LABLOG

"Celsius or Fahrenheit?"

No inventing.

YOU SAY

"What chemical next?"

LABLOG

Answers only from the approved protocol. Not there?
"Please verify first."

No overwriting.

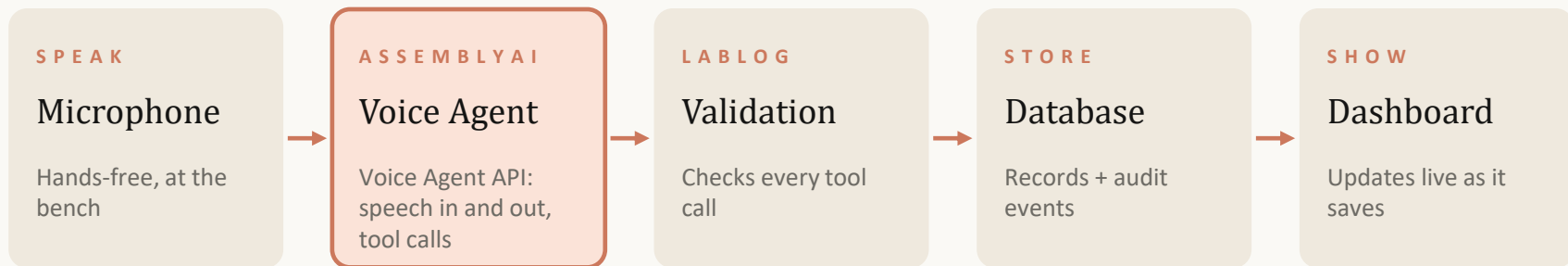
YOU SAY

"Actually, make it 4.3."

LABLOG

4.3 °C saved. Previous 4.2 stays in the audit trail.

One live voice loop. Typed tools. Validated writes.



WHY ASSEMBLYAI IS CENTRAL

- **One loop.** Listening, reasoning, speaking and tool calls over a single WebSocket.
- **See what it heard.** Partial and final transcripts stream to the screen.
- **Interrupt it.** Barge-in: cut it off mid-answer and it stops and listens.
- **Lab words.** Measurement names, units and sample IDs bias recognition.

We test the agent like software.

0

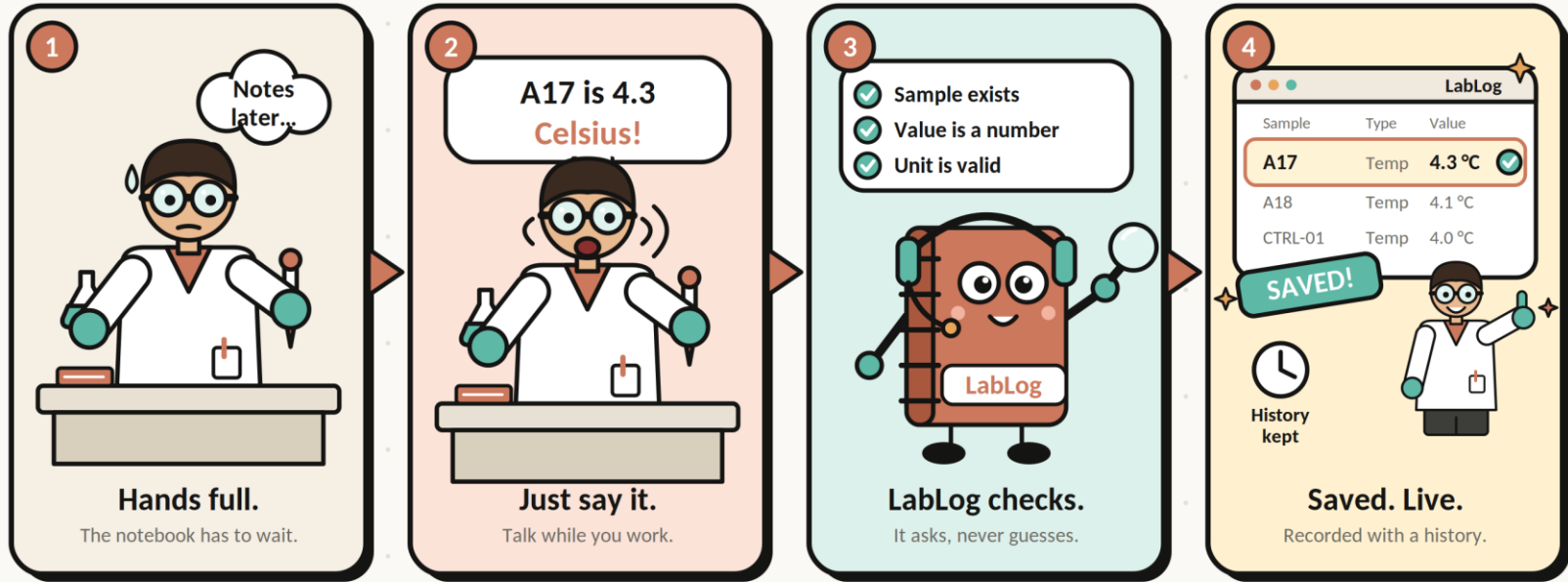
procedures invented.
The target we hold ourselves to.

SCRIPTED VOICE SCENARIOS

Normal reading	correct tool call
Missing sample	asks who it belongs to
Correction	correction, not a new reading
Unknown sample	backend rejects it
Interruption	agent stops and listens
Unsupported procedure	declines to invent
Finish with gaps	completeness check first

Also targeting: 100% of ambiguous critical actions clarified, 100% completion checks, >95% measurement recognition.

Hear it. Check it. Save it.



Asks, never guesses • Corrections keep history • Answers only from the approved protocol

Built for the lab. Ready to grow.

NOW

- Voice logging of readings, steps and events
- Protocols, deviations, corrections
- Live dashboard and audit trail

NEXT

- Scan a barcode, then just speak
- Compare with earlier runs
- Notify teammates on Slack or email

LATER

- SOP PDF import, approved by a person
- Instrument data ingestion
- Integration with existing ELN and LIMS

Who it's for: biotech, pharma R&D, university, food-testing and QC labs. **How it grows:** a voice layer for existing lab software, from \$49 per user per month.

Say it. Log it. Trust it.

Original

A lab workflow system controlled by voice.

Technically deep

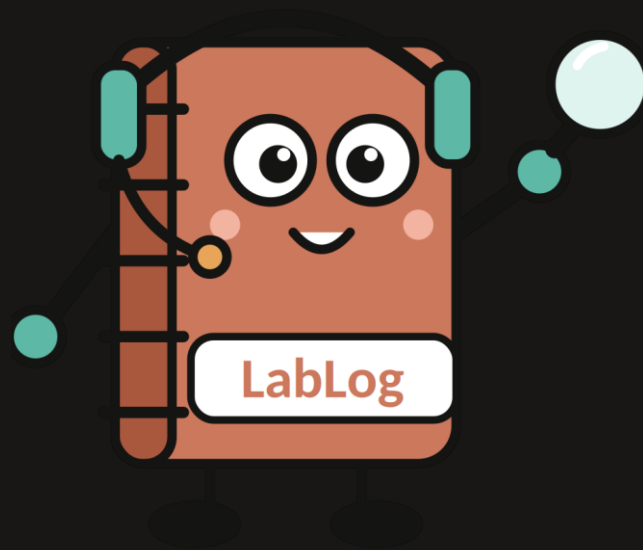
Live voice loop, typed tools, validated writes.

Trustworthy

Asks, never guesses. Never invents.

Commercial

A voice layer for existing lab software.



Run experiments with your hands. Document them with your voice.